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Current-voltage (I-V) characteristics of Au/n-Ge heterostructure based on cobalt phthalocyanine (CoPc) interlayer

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ABSTRACT

The influence of cobalt phthalocyanine (CoPc) interlayer on the I-V measurements of Au/n-Ge (pristine) contacts is investigated. I-V characteristics reveals the decrease in the reverse leakage current of CoPc interlayer contacts than compared to contacts without interlayer. The Schottky barrier parameters were described using forward I-V measurements by assuming standard thermionic emission mechanism. The barrier height (BH) of the contact with CoPc interlayer is significantly improved than compared to pristine contact. The observed BHs of pristine and CoPc interlayer contacts are found to be 0.63 eV and 0.67 eV respectively. Reduced interface state density and altered space charges on the n-Ge surface reduces tunneling resistance, may account for the larger BH observed in contacts with CoPc interlayer than pristine contacts. The series resistance evaluated for the contacts using Cheung's method shows its consistency with Norde method. The observed series resistance value of the Au/CoPc/n-Ge contacts found to be significantly smaller in magnitude than Au/n-Ge contacts. The current conduction mechanism of pristine contact and contact with interlayer was evaluated using double logarithmic I-V plot in the forward bias region. Three regions of conduction mechanisms are observed for Au/CoPc/n-Ge contacts. It is evident that the CoPc interfacial layer considerably changes the conduction mechanism from ohmic conduction to trap charge limited current (TCLC) mechanism at moderately high voltages. The associated transition in the conduction mechanism might be due to the presence of shallow traps (low energy traps) with exponential distribution at the CoPc/n-Ge interface.

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1. Introduction

Ge semiconductor is found to be a good alternative for the fabrication of high mobility CMOS devices than compared with Si [1]. Metal-Ge Schottky contacts have recently regained popularity for the following reasons: (1) High speed Ge-based electronic devices are possible due to its enhanced magnitudes of electron and hole mobility in low electric fields against Si semiconductor technology [2], (2) Due to its compatibility with Si, the main component of VLSI technology is Ge [3]. Before Ge can take the lead in cutting-edge CMOS technology, a few challenges must be overcome. Pinning of Fermi-level close to the valence band maximum of Ge when metals come into contact with n-Ge hinders the effective Schottky barrier height (SBH) [4]. An interfacial layer such as conducting polymers, ceramic materials, insulators or oxide layers modifies the surface of the semiconductor which reduces the metal induced surface states and interface state density which further improves the quality of the rectification properties of the contact. Acceptor based organic semiconductors are potentially being studied due to their application in electrical devices like OFETs, bulk heterojunction solar cells, LEDs, solar cells and gas sensors.

Metal phthalocyanines, on the other hand, have grown its importance in the development of organic based optoelectronic devices due to their unique properties, which include ease of fabrication, high flexibility, low molecular weight, improved electrical properties, wide light absorption, and a very high molecular coefficient [5]. These materials have demonstrated good chemical and thermal stability. One such significant organic semiconductor

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is cobalt phthalocyanine, $C_{32}H_{16}CoN_8$ (CoPc). Due to its chemical and biological properties, extensive research was carriedout in the last few decades to understand its suitability for various opto-electronics devices [6]. Pei-Te Lin et al. fabricated Pt/Ge contact using inexpensive interlayer such as carbon paste (CP) and studied its electrical properties. The Pt/CP/Ge contact exhibits improved rectifying behavior and shows higher barrier height than conventional contact without interlayer [7]. Sunil Babu et al. developed NiGe/p-Ge contacts using an ultra-low resistance Terbium (Tb) interlayer [8]. The Accumulation of Tb atoms at the interface causes to increase the barrier width which in turn decreases the barrier height of the contacts. Iksoo Park et al. demonstrated the electrical measurements and high temperature stability of Ti/TiO_2/n-Ge contacts for different carbon incorporation [9]. The MIS contacts were found to be degraded at 550 °C annealing temperatures due to the deformation of TiO_2 interlayer. Further, it is noted that the thermal stability of MIS contacts were highly improved by carbon incorporation and evidenced the enhanced performance of Ge-based devices. α-phase CuPc nanoparticles were synthesized using solution method and the electrical measurements of the CuPc modified n-Si contact has been analysed [5]. The device shows good rectification ratio and investigation of photocurrent sensitivity measurements under various illumination intensities suggest the existence of traps in deep levels in the organic material.

In the present work, a CoPc modified n-Ge contact was fabricated and analysed its current–voltage measurements. It is very important to investigate the various current conduction mechanisms influencing the device properties. Further the low series resistance, high barrier height and low reverse leakage current observed in the case of n-Ge contacts modified with CoPc interlayer shows its applicability in various optoelectronic and photovoltaic devices. Further this work can be extended by studying the electrical properties of n-Ge contacts with CoPc interlayer using different rapid thermal annealing temperatures and also by aging method.

2. Device fabrication procedure

In the present work, an n-type Ge (100) (Sb-doped) wafer was employed with a carrier concentration of $2x10^{18}$ cm^{-3}. Initially semiconductor samples were ultrasonicated in H_2O:HF (100:1) solution to remove the oxide layer on the n-Ge surface. CoPc thin layer with 50 nm thickness was formed on the cleaned Ge wafer using vacuum evaporation (evidenced from profilometer). Specifically, a 0.5 mm diameter shadow metal mask was used to form 40 nm-thick Schottky metal (Au) on top of the CoPc thin film with e-beam evaporation technique. By using the same procedure, a reference (pristine) sample is also fabricated without interfacial layer. A precision Agilent 4155C based I-V source meter and a precision Agilent 4284A LCR meter were used to measure the electrical properties of the CoPc interlayer contacts and pristine samples at room temperature.

3. Results and discussion

I-V measurements of the Au/n-Ge contacts with and without CoPc interlayer is depicted in Fig. 1. Superior barrier characteristics such as small reverse leakage currents, reduced series resistance and large BH values are observed for contacts with CoPc interlayer than compared to pristine sample. The performance of the Schottky contacts were evaluated for small forward bias voltages ($V > 3kT/q$) by assuming standard thermionic emission mechanism. The ideality factor and SBH are found to be 0.63 eV, 1.11 and 0.67 eV, 1.58 for the pristine contact and CoPc interlayer contacts respectively. The ideality factor of the pristine and CoPc interlayer contacts found significantly higher than unity, might be explained based on the presence of dipoles at the interface by CoPc interlayer or a particular interface structure, as well as flaws brought on by fabrication processes and inhomogeneous barrier at the interface [10]. Moreover, the contacts with CoPc interlayer shows a larger SBH than pristine contact. It is well known that tunnel resistance found to be reduced by depinning of Fermi level at the Ge-surface using a thin interfacial layer which further helps to modify SBH of the MIS contacts. Nevertheless, the addition of a moderately thick CoPc interlayer affect the space charge region, changing the effective barrier height [11]. Alternatively, it is assumed that the CoPc layer causes a significant shift in the semiconductors electron affinity and the metals work function. It is obvious that the Au/CoPc/n-Ge Schottky diode exhibits non-ideal characteristics with high ideality factor as a result of the existence of a CoPc interlayer. Interface states, series resistance and the existence of an interfacial layer are a few factors that lead to deviations from ideal behavior of the contacts [12].

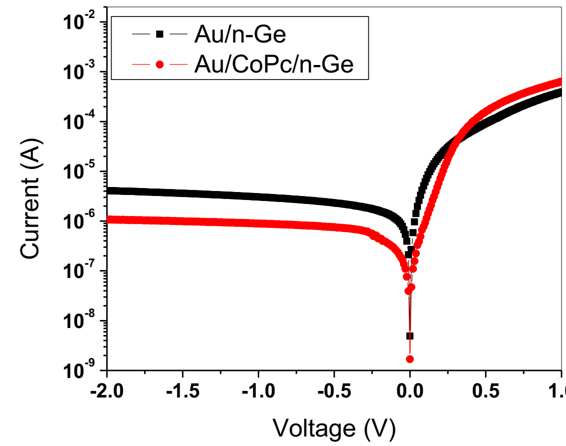

Fig. 1. I-V characteristics of pristine and CoPc interlayer contacts.

The barrier parameters cause the deviation from the non-linear I-V characteristics preferably at higher forward bias voltages. In such circumstances, Cheung's approach can be used to exactly compute the barrier parameters at significantly high current ranges. The R_S and n values for the pristine contact are found to be 3027 Ω and 1.97 respectively, and 981 Ω and 2.34 for the contacts with CoPc interlayer which are evaluated from dV/dlnI versus V plot as shown in Fig. 2a and b.

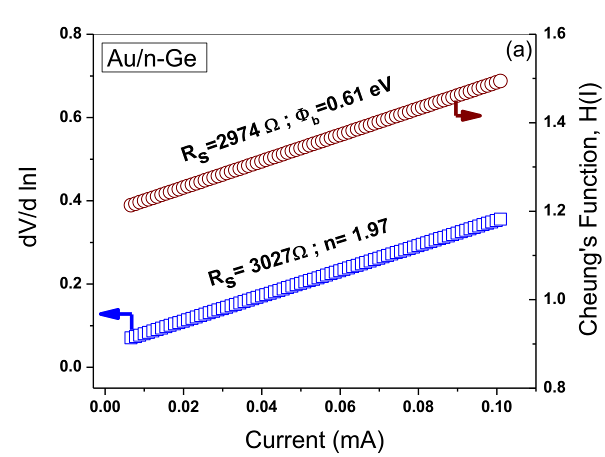

Fig. 2a. Cheung's plots of pristine contacts.

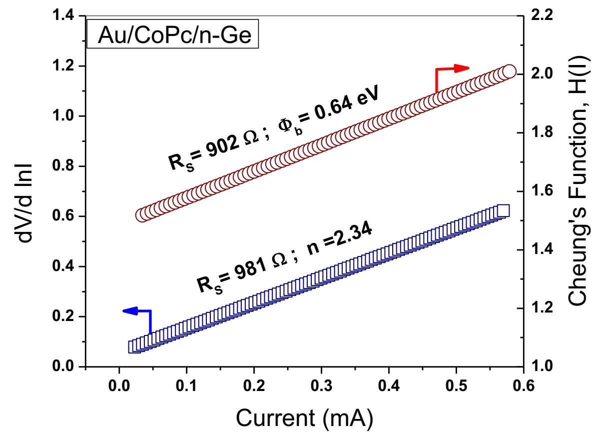

Fig. 2b. Cheung's plots of CoPc interlayer contacts.

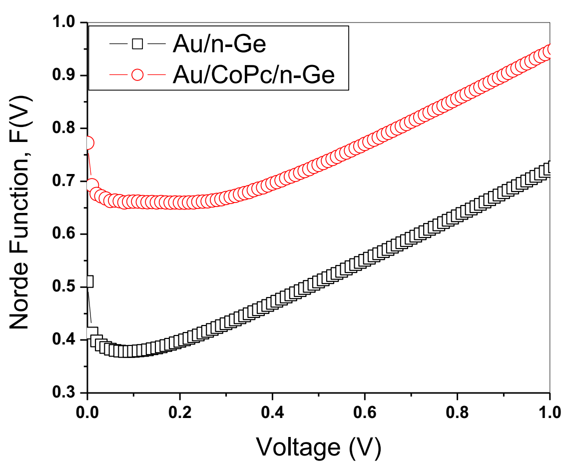

Fig. 3. Norde plot of pristine contacts and CoPc interlayer contacts.

According to the H(I) versus I plot, R_S and SBH values for the pristine and CoPc interlayer contacts are evaluated as 2974 Ω and 0.61 eV and 902 Ω and 0.64 eV respectively (Fig. 2a and b). A significant difference in ideality factor values observed in ln I-V plot and cheung's plot for the contacts with and without CoPc interlayer. This deviation might be due to the voltage drop across the interfacial layer, series resistance, presence of interface states, and dependency of SBH with bias [13]. From this observation, it is evident that the CoPc interlayer modifies the structure of the interface and reduced the series resistance of the pristine contact. Fig. 3 shows the plot of Norde function F(V) vs. voltage of Au/n-Ge contacts with and without CoPc interlayer. The SBH and R_S values were evaluated for pristine sample found to be 0.63 eV and 2458 Ω and 0.73 eV and 87.4 Ω for the CoPc interlayer contact respectively. The Schottky barrier parameters evaluated by different approaches is mentioned in Table 1.

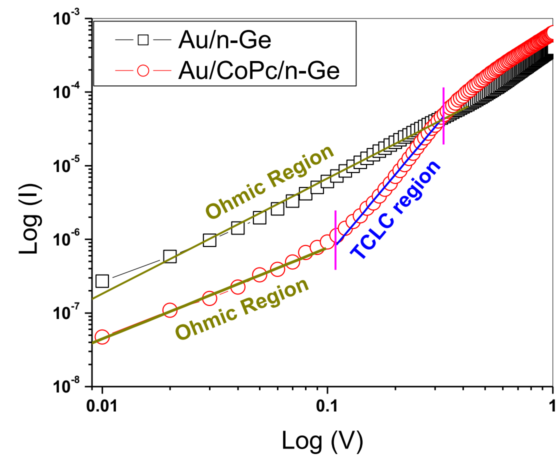

Fig. 4. Au/n-Ge and Au/CoPc contacts current-conducting mechanisms in the Forward I-V measurements.

Fig. 4 illustrates the current conduction mechanisms of pristine contacts and contacts with CoPc interlayer using double logarithmic scale of forward I-V measurements. At different bias regions the current conduction mechanisms are determined using slope factor ('m') obtained by fitting the I-V data linearly ($I \alpha V^m$). Three linear regions were observed in Au/CoPc/n-Ge contacts with different slope factors. In the case of Au/CoPc/n-Ge Schottky contacts, a transition from ohmic conduction mechanism to trap assisted space charge limited current (TCLC) is observed at higher bias regions. This might be due to the traps or defects observed at the interface below the conduction band of n-Ge semiconductor. As a result, traps or defects at the CoPc interlayer considerably control the current conduction in Au/CoPc/n-Ge contacts [14]. The presence of traps or defects at the organic–inorganic interface can further be decreased by aging, annealing the contacts and by using different surface passivation methods before the deposition of organic thin films [15–16].

4. Conclusions & future scope

CoPc interlayer preferably influences the current–voltage characteristics of the pristine contacts. The Schottky contacts shows good rectification behavior and significantly high rectification ration in the case of contacts with CoPc interlayer.

- Notably, large SBH and ideality factor (n > 1) observed in the case of Au/CoPc/n-Ge devices than pristine samples.
- The SBH value of CoPc contacts (0.67 eV) was found to be much higher than the typical Au/n-Ge MS contact (0.63 eV).
- An organic CoPc interlayer has been used to modify the barrier properties and space region of the metal/n-Ge contacts.
- Due to the formation of defects or traps at the active CoPc layer of the contact below the conduction band, contacts with interlayer exhibits a trap charge-limited current conduction at higher bias range.

Table 1
Schottky barrier properties of pristine and with CoPc interlayer contacts evaluated through various techniques.

Sample	Forward I-V		dV/d(lnI) vs I		H(I) vs I		Norde	
	Φ_b (eV)	n	Rs (Ω)	n	Rs (Ω)	Φ_b (eV)	Rs (Ω)	Φ_b (eV)
Au/n-Ge	0.63 eV	1.51	3027 Ω	1.97	2974 Ω	0.61 eV	2458 Ω	0.63 eV
Au/CoPc/n-Ge	0.67 eV	1.58	981 Ω	2.34	902 Ω	0.64 eV	87.4 Ω	0.73 eV

- Further this work can be extended by studying the electrical properties of n-Ge contacts with CoPc interlayer using different rapid thermal annealing temperatures and also by aging method.
- Further the low series resistance, high barrier height and low reverse leakage current observed in the case of n-Ge contacts modified with CoPc interlayer shows its applicability in various optoelectronic and photovoltaic devices.

CRediT authorship contribution statement

M. Pavani: Conceptualization, Methodology, Data curation, Formal analysis, Writing – original draft. **A. Ashok Kumar:** Conceptualization, Methodology, Data curation, Validation, Writing – review & editing. **V. Rajagopal Reddy:** Conceptualization, Methodology, Validation, Writing – review & editing. **V. Janardhanam:** Resources, Data curation, Validation. **Chel-Jong Choi:** Resources, Data curation, Validation.

Data availability

Data will be made available on request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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